

A Short Guide to Lemma C.2

Compact-observable limits of completely positive maps

Abstract

Lemma C.2 is the compactness device used in the unified Gaussian converse proof. It extracts a completely positive limit from a uniformly bounded sequence of completely positive maps, even though trace-norm compactness is unavailable in infinite dimension. Convergence is tested only with compact output observables. This note explains the choice of topology, the diagonal argument on the adjoints, the possible loss of trace, and the passage from asymptotic to exact covariance.

1 What the lemma does

For a Hilbert space $\mathcal{H}$, we use the operator classes

$$\begin{aligned}\mathcal{B}(\mathcal{H}) &:= \{A : \mathcal{H} \rightarrow \mathcal{H} : A \text{ is linear and bounded}\}, \\ \mathcal{K}(\mathcal{H}) &:= \overline{\{A \in \mathcal{B}(\mathcal{H}) : \text{rank } A < \infty\}}^{\|\cdot\|}, \\ \mathcal{T}_1(\mathcal{H}) &:= \{X \in \mathcal{K}(\mathcal{H}) : \text{Tr } |X| < \infty\}, \quad \|X\|_1 := \text{Tr } |X|,\end{aligned}$$

where $|X| = (X^*X)^{1/2}$. Thus $\mathcal{B}(\mathcal{H})$, $\mathcal{K}(\mathcal{H})$, and $\mathcal{T}_1(\mathcal{H})$ are respectively the bounded, compact, and trace-class operators, with $\mathcal{T}_1(\mathcal{H}) \subset \mathcal{K}(\mathcal{H}) \subset \mathcal{B}(\mathcal{H})$.

Let

$$\Gamma_j : \mathcal{T}_1(\mathcal{H}_0) \longrightarrow \mathcal{T}_1(\mathcal{H}_1)$$

be completely positive maps that are uniformly bounded as maps between the two trace-class spaces: for some $M < \infty$,

$$\sup_j \|\Gamma_j\|_{1 \rightarrow 1} := \sup_j \sup_{X \neq 0} \frac{\|\Gamma_j(X)\|_1}{\|X\|_1} \leq M.$$

For each fixed X , this bound only places the outputs $\Gamma_j(X)$ in a bounded subset of the trace class, and such subsets are not generally precompact in trace norm in infinite dimension (orthogonal rank-one states, for example, remain pairwise trace distance two). Lemma C.2 instead uses the weaker topology obtained by testing outputs against compact operators. Since $\mathcal{T}_1(\mathcal{H}_1) = \mathcal{K}(\mathcal{H}_1)^*$, Banach–Alaoglu makes every closed norm-bounded ball weak-* compact; moreover, separability of $\mathcal{K}(\mathcal{H}_1)$ makes this topology metrizable on such a ball, so every bounded sequence has a weak-* convergent subsequence. This allows one to extract a map Γ such that

$$\text{Tr}[\Gamma_j(X)K] \longrightarrow \text{Tr}[\Gamma(X)K] \tag{1}$$

for every trace-class input X and every compact output observable K .

This is exactly the convergence needed later. The quantum witness W_γ appearing in Proposition C.5 is compact, so its moment passes to the limit by (1).

The lemma also records two stability properties:

- a common trace bound passes to the limit, although equality of traces need not;
- if the maps become covariant in trace norm, the limiting map is exactly covariant.

2 The topology behind the statement

2.1 Trace class is the dual of the compact operators

For a separable Hilbert space $\mathcal{H}$,

$$\mathcal{K}(\mathcal{H})^* = \mathcal{T}_1(\mathcal{H}), \quad \langle X, K \rangle = \text{Tr}(XK).$$

Consequently, bounded sets of trace-class operators are compact in the weak-* topology induced by compact observables. A sequence X_j converges to X in this topology precisely when

$$\text{Tr}(X_j K) \longrightarrow \text{Tr}(XK) \quad (K \in \mathcal{K}(\mathcal{H})).$$

This topology is weaker than the usual weak topology obtained by testing against all bounded operators. In particular, the identity is not compact when $\mathcal{H}$ is infinite-dimensional, so weak-* convergence on $\mathcal{K}(\mathcal{H})$ does not force convergence of traces.

For example, if (e_n) is an orthonormal basis, then

$$|e_n\rangle\langle e_n| \xrightarrow{w^*} 0 \quad \text{on } \mathcal{K}(\mathcal{H}),$$

where w^* is read “weak star” and means convergence after pairing with every compact operator. Indeed,

$$\text{Tr}(|e_n\rangle\langle e_n|K) = \langle e_n, K e_n \rangle \longrightarrow 0 \quad (K \in \mathcal{K}(\mathcal{H})),$$

even though every operator $|e_n\rangle\langle e_n|$ has trace one. Compact observables cannot see a unit amount of mass moving to arbitrarily high basis vectors. This is the “escape to infinity” allowed by Lemma C.2.

2.2 Why the proof works with adjoints

The Banach adjoint of Γ_j is

$$\Gamma_j^* : \mathcal{B}(\mathcal{H}_1) \longrightarrow \mathcal{B}(\mathcal{H}_0), \quad \text{Tr}[\Gamma_j(X)A] = \text{Tr}[X\Gamma_j^*(A)].$$

Restricting Γ_j^* to $\mathcal{K}(\mathcal{H}_1)$ turns the problem into extracting a pointwise weak-* limit in $\mathcal{B}(\mathcal{H}_0) = \mathcal{T}_1(\mathcal{H}_0)^*$. Banach–Alaoglu gives the required compactness on bounded sets of this dual space.

3 Statement of Lemma C.2

Lemma C.2 (rephrased). *Let $\mathcal{H}_0, \mathcal{H}_1$ be separable Hilbert spaces and let $\Gamma_j : \mathcal{T}_1(\mathcal{H}_0) \rightarrow \mathcal{T}_1(\mathcal{H}_1)$ be completely positive maps whose induced norms $\|\Gamma_j\|_{1 \rightarrow 1}$ are uniformly bounded. Then a subsequence and a completely positive map Γ exist such that*

$$\text{Tr}[\Gamma_j(X)K] \longrightarrow \text{Tr}[\Gamma(X)K]$$

for all $X \in \mathcal{T}_1(\mathcal{H}_0)$ and $K \in \mathcal{K}(\mathcal{H}_1)$.

If, for every positive X ,

$$\mathrm{Tr} \Gamma_j(X) \leq c \mathrm{Tr} X,$$

then $\mathrm{Tr} \Gamma(X) \leq c \mathrm{Tr} X$. Suppose U_g on $\mathcal{H}_0$ and V_g on $\mathcal{H}_1$ are families of unitaries indexed by a set G and satisfy

$$\|\Gamma_j(U_g X U_g^*) - V_g \Gamma_j(X) V_g^*\|_1 \rightarrow 0$$

for every fixed g and X , then

$$\Gamma(U_g X U_g^*) = V_g \Gamma(X) V_g^*.$$

4 Proof walkthrough

Step 1: diagonalize on a countable family of observables

Because $\mathcal{H}_1$ is separable, $\mathcal{K}(\mathcal{H}_1)$ is separable in operator norm. Choose a countable norm-dense $\mathbb{Q}(i)$ -linear $*$ -algebra $\mathcal{D} \subset \mathcal{K}(\mathcal{H}_1)$, where $\mathbb{Q}(i) = \{a + ib : a, b \in \mathbb{Q}\}$ is the countable dense subfield of Gaussian rationals in $\mathbb{C}$. Using this field keeps $\mathcal{D}$ countable while preserving addition, products, adjoints, and enough scalar multiplication to extend the limiting map uniquely to a complex-linear map by continuity.

For every fixed $K \in \mathcal{D}$, the operators $\Gamma_j^*(K)$ form a bounded subset of $\mathcal{B}(\mathcal{H}_0)$. Banach–Alaoglu gives a weak- $*$ convergent subsequence because the predual $\mathcal{T}_1(\mathcal{H}_0)$ is separable and the bounded weak- $*$ compact sets are therefore metrizable. Applying this successively to the countably many elements of $\mathcal{D}$ and taking the diagonal subsequence gives

$$\Gamma_j^*(K) \xrightarrow{w^*} \Gamma^*(K) \quad (K \in \mathcal{D}).$$

The common norm bound makes the limiting assignment bounded. It therefore extends uniquely from $\mathcal{D}$ to a bounded complex-linear map

$$\Gamma^* : \mathcal{K}(\mathcal{H}_1) \rightarrow \mathcal{B}(\mathcal{H}_0), \tag{2}$$

and the same bound extends convergence to every compact K .

Step 2: complete positivity survives the limit

Complete positivity must be checked at every matrix level. Fix $A \geq 0$ in $M_r(\mathcal{K}(\mathcal{H}_1))$. The square root $A^{1/2}$ is compact. Choose $B_\ell \in M_r(\mathcal{D})$ with $B_\ell \rightarrow A^{1/2}$ in norm. Complete positivity of every Γ_j^* gives

$$\Gamma_j^{*(r)}(B_\ell^* B_\ell) \geq 0.$$

First let $j \rightarrow \infty$ and then let $\ell \rightarrow \infty$. The positive cone is weak- $*$ closed, so $\Gamma^{*(r)}(A) \geq 0$. Since this holds for every r , the map in (2) is completely positive.

Step 3: return to a map on trace class

For $X \in \mathcal{T}_1(\mathcal{H}_0)$, define a functional on $\mathcal{K}(\mathcal{H}_1)$ by

$$K \mapsto \mathrm{Tr}[X \Gamma^*(K)].$$

It is bounded, hence it is represented by a unique trace-class operator $\Gamma(X)$ because $\mathcal{K}(\mathcal{H}_1)^* = \mathcal{T}_1(\mathcal{H}_1)$. Thus

$$\mathrm{Tr}[\Gamma(X)K] = \mathrm{Tr}[X \Gamma^*(K)]. \tag{3}$$

The complete positivity of Γ^* implies complete positivity of Γ . Combining the construction of Γ^* with (3) proves the desired convergence (1).

Step 4: recover the trace inequality

Let P_N be increasing finite-rank projections on $\mathcal{H}_1$ with $P_N \uparrow I$ strongly. For a positive trace-class operator Y ,

$$\mathrm{Tr} Y = \sup_N \mathrm{Tr}(Y P_N).$$

Therefore, for $X \geq 0$,

$$\begin{aligned} \mathrm{Tr} \Gamma(X) &= \sup_N \mathrm{Tr}[\Gamma(X) P_N] \\ &= \sup_N \lim_j \mathrm{Tr}[\Gamma_j(X) P_N] \\ &\leq \liminf_j \mathrm{Tr} \Gamma_j(X) \\ &\leq c \mathrm{Tr} X. \end{aligned}$$

The third line uses $\sup_N \lim_j a_{j,N} \leq \liminf_j \sup_N a_{j,N}$ with $a_{j,N} = \mathrm{Tr}[\Gamma_j(X) P_N]$; here $\sup_N a_{j,N} = \mathrm{Tr} \Gamma_j(X)$ because $\Gamma_j(X) \geq 0$ and $P_N \uparrow I$. Thus the trace bound passes to the limit even though the traces themselves need not converge. The argument gives an inequality rather than equality because the identity is approximated from below only after the limit has been taken. Escaped mass contributes to $\mathrm{Tr} \Gamma_j(X)$ but to none of the fixed compact projections P_N .

Step 5: asymptotic covariance becomes exact

Fix g , X , and $K \in \mathcal{K}(\mathcal{H}_1)$. Pair the covariance defect with K :

$$\begin{aligned} &\left| \mathrm{Tr} \left[(\Gamma_j(U_g X U_g^*) - V_g \Gamma_j(X) V_g^*) K \right] \right| \\ &\leq \| \Gamma_j(U_g X U_g^*) - V_g \Gamma_j(X) V_g^* \|_1 \|K\|. \end{aligned}$$

The right side tends to zero. On the second term, cyclicity replaces K by $V_g^* K V_g$, which is still compact. Passing to the limit gives

$$\mathrm{Tr}[\Gamma(U_g X U_g^*) K] = \mathrm{Tr}[V_g \Gamma(X) V_g^* K].$$

Compact operators separate trace-class operators, so the two outputs are equal. This proves exact covariance for every fixed g .

5 Points worth keeping straight

1. The lemma does not assert trace-norm convergence of outputs. It asserts convergence only after testing with a compact observable.
2. A sequence of channels may converge to a trace-nonincreasing map. This is a genuine infinite-dimensional phenomenon, not a defect in the proof.
3. Complete positivity is not obtained from positivity alone; the closed-cone argument must be made at every matrix level.
4. The covariance hypothesis is pointwise in g and X . That is enough because the conclusion is also proved one fixed g at a time.

6 Role in Appendix C

In the orbital specialization of Proposition C.5, Følner-averaged channels are asymptotically displacement covariant. Lemma C.2 extracts an exactly covariant trace-nonincreasing limit, and compactness of W_γ lets its weighted moment pass to that limit. In the full classical–quantum specialization, the same lemma is applied to the weighted quantum maps Γ_j^χ . Their initial trace bound is only $\|\chi\|_\infty$; the classical convolution argument later improves the limit to a trace-nonincreasing map. Thus Lemma C.2 supplies the quantum compactness and exact covariance common to both cases of the unified proposition.